Report

Of

SQL and Database Management Systems

(AICTE – EduSkills Virtual Internship)

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

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CANDIDATE'S DECLARATION

I, Varun Tomar do hereby declare that the internship report titled **"SQL and Database Management Systems"** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material

from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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I express my gratitude to my parents for their blessings.

Signature

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INTERNSHIP COMPLETION CERTIFICATE



PROJECT DESCRIPTION

4.1 Introduction

Data is at the core of every modern software system, and the ability to design, store, query, secure and analyse it is a fundamental skill for a computer science graduate. Internships bridge the gap between academic knowledge and real-world applications by giving hands-on exposure to industry-relevant tools and practices.

This report presents the 8-week AICTE – EduSkills Virtual Internship in the domain of SQL and Database Management Systems. The internship covered

relational database design and SQL, NoSQL databases (MongoDB), data aggregation and ETL, data visualization with Power BI, integration of databases with full-stack applications (Node.js & Express and Python & FastAPI), database performance, security and backup, scalability and high availability, and a capstone project on building a data-driven business analytics dashboard.

4.2 Organization Profile

Organization Name: EduSkills Academy (EduSkills – "Nation Building Through Skills")

Domain: EdTech & Skill Development

Services: Virtual Internships and industry-oriented skill programmes

Supported by: AICTE (All India Council for Technical Education), under the patronage of the National Internship Portal

Recognition: ISO-certified organization and Great Place To Work certified

Internship Overview

- Internship Mode: Online / Virtual
- Domain: SQL and Database Management Systems
- Duration: 8 Weeks (April 2026 – June 2026)
- Platform Used: EduSkills virtual internship platform (AICTE – EduSkills)
- Role: Intern
- Offer Letter Ref. No.: C16Y2026M051441098
- AICTE Student ID: STU68346ab0a07001748265648

4.3 Problem Statement

Modern organizations generate large volumes of data from sales, customers, applications and web services. To make use of this data, it must be stored in well-designed databases, retrieved efficiently, protected against misuse and loss, integrated with applications through APIs, and finally presented as meaningful insights for decision-making. Students often learn these topics in isolation, without seeing how a relational database, a NoSQL store, an ETL process, a backend API and a reporting tool work together.

This internship addresses that gap through a structured, week-wise programme that moves from database fundamentals to advanced querying, data integration, performance, security and scalability, and ends with a capstone project: building a data-driven business analytics dashboard.

4.4 Project Objectives

- To understand database concepts and design normalized relational databases using ER modeling and SQL (DDL and DML).
- To write advanced SQL queries using joins, subqueries, aggregation, GROUP BY and HAVING.
- To understand NoSQL systems and perform CRUD and aggregation operations in MongoDB.
- To apply ETL and data transformation techniques and visualize data using Power BI (data modeling, relationships and basic DAX).
- To integrate databases with full-stack applications and build RESTful CRUD APIs using Node.js & Express and Python FastAPI with SQLAlchemy.
- To apply performance optimization, security (roles, permissions, SQL injection prevention) and backup strategies.
- To understand scalability and high-availability concepts such as replication, sharding and clustering.
- To build a data-driven business analytics dashboard as the capstone project.

4.5 Scope of the Project

The scope of the internship covers the complete data lifecycle: designing the database, loading and transforming data, exposing it through APIs, securing and tuning it, and presenting it in an analytics dashboard. The learning can be extended in the future in the following directions:

- Working with larger, real-time datasets and cloud-hosted database services.
- Adding automated ETL pipelines and scheduled data refresh for the dashboard.
- Implementing replication, sharding and automated backup in a production-like environment.
- Extending the analytics with advanced DAX measures, time-series forecasting and geospatial analysis.

Technologies and Tools Used

Relational Databases: SQL,PostgreSQL – DDL, DML, joins, subqueries, aggregation, indexing, roles and permissions.

NoSQL Database: MongoDB – document model, CRUD operations, aggregation pipeline, advanced and geospatial indexing, time series data modeling.

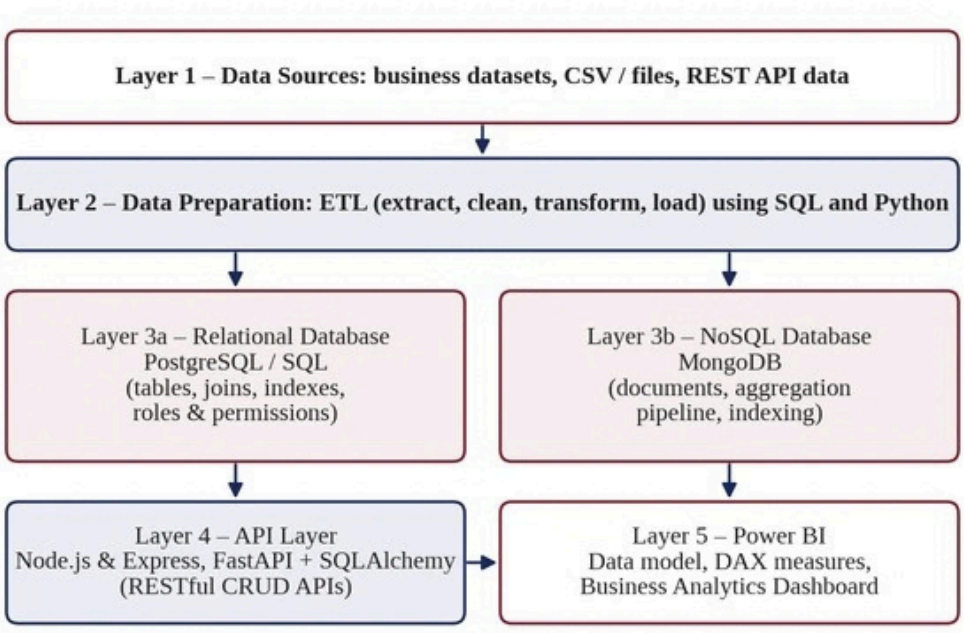
Data Analytics & Visualization: ETL techniques, Power BI (data connections, model view, relationships, basic DAX, dashboards).

Backend & Integration: Node.js and Express; Python with FastAPI and SQLAlchemy; RESTful CRUD APIs; API consumption using the requests library.

Concepts: ER modeling, normalization, query optimization, SQL injection prevention, backup and recovery, replication, sharding, clustering, high availability.

System Architecture

The capstone solution follows a layered architecture. Data from business datasets and API sources is prepared through ETL, stored in relational (PostgreSQL) and NoSQL (MongoDB) databases, exposed through REST APIs built with Node.js/Express and FastAPI, and finally modelled and visualized in Power BI as a business analytics dashboard.



Layered architecture of the data-driven analytics solution

Methodology

The internship followed a week-wise structured learning plan. Each week combined concept learning with practical exposure, weekly assessments were used to track progress, certain modules required project documentation and assignments, and the final week included a Final Assessment Test. The final

grade was based on overall performance across weekly assessments, project submissions and the final test.

Week-wise structure of the internship

Week	Module	Description
1	Database Foundations & Relational Design	Introduction to database concepts, SQL vs NoSQL systems, environment setup, relational database design, ER modeling, normalization, DDL and basic DML operations (SELECT, INSERT, UPDATE, DELETE).
2	Advanced SQL & NoSQL Fundamentals (MongoDB)	Advanced SQL queries including joins, subqueries, aggregation, GROUP BY/HAVING; introduction to NoSQL concepts, MongoDB document model, and CRUD operations.
3	Data Aggregation, ETL & Power BI Fundamentals	Advanced SQL aggregation, ETL concepts, data transformation techniques, introduction to the Power BI interface, data connections, and building basic dashboards and visualizations.
4	Data Modeling & Full-Stack Database Integration (Node.js)	Power BI model view, relationships, basic DAX, performance optimization; integrating databases with full-stack applications using Node.js & Express, PostgreSQL connection, and CRUD API development.
5	Backend APIs & Data Integration (FastAPI)API consumption using	FastAPI with SQLAlchemy, building RESTful CRUD APIs, the requests library, storing API data, and combining multiple datasets.
6	Database Performance, Security & Backup	Performance metrics, indexing strategies, query optimization, configuration tuning, user roles & permissions, SQL injection prevention, backup and recovery strategies.
7	Scalability, High Availability & Real-World Use Cases	Scaling strategies, replication, sharding, clustering concepts, case studies from e-commerce, social media and financial services, and consolidated best practices.
8	Advanced NoSQL, Time Series Data Modeling & Capstone Project	MongoDB aggregation pipeline, advanced indexing, geospatial queries, time series data modeling, data quality, reproducibility, model validation and deployment concepts; Capstone Project and Internship Grade Point Assessment.

Capstone Project: Data-Driven Business Analytics Dashboard

The capstone brought the individual modules together. Business data was extracted and cleaned using SQL and ETL techniques, stored in databases,

combined with other datasets (including data consumed from APIs), and loaded into Power BI. A data model with relationships was created, DAX measures were defined, and interactive visuals were arranged into a dashboard that presents business performance for decision-making. Advanced NoSQL querying with the MongoDB aggregation pipeline complemented the relational analysis.

Illustrative Examples of Concepts Practised

The following short examples illustrate the kinds of queries and code used while learning the above modules.

```
-- Relational design (normalized tables, DDL)
CREATE TABLE customers (

  customer_id SERIAL PRIMARY KEY,
  name        VARCHAR(100) NOT NULL,
  email       VARCHAR(120) UNIQUE
);
CREATE TABLE orders (
  order_id    SERIAL PRIMARY KEY,
  customer_id INT REFERENCES customers(customer_id),
  order_date DATE,
  amount      NUMERIC(10,2)
);


-- Join with aggregation (GROUP BY / HAVING)
SELECT c.name, COUNT(o.order_id) AS total_orders,
       SUM(o.amount) AS total_spent
FROM customers c
JOIN orders o ON o.customer_id = c.customer_id
GROUP BY c.name
HAVING SUM(o.amount) > 10000
ORDER BY total_spent DESC;


-- Performance: index + query plan
CREATE INDEX idx_orders_customer ON orders(customer_id);
EXPLAIN ANALYZE
SELECT * FROM orders WHERE customer_id = 42;

--Security: least-privilege role
CREATE ROLE analyst LOGIN PASSWORD '********';
GRANT SELECT ON customers, orders TO analyst;


// Node.js + Express: parameterized query (prevents SQL injection)
app.get('/customers/:id', async (req, res) => {
  const result = await pool.query(
    'SELECT * FROM customers WHERE customer_id = $1',
    [req.params.id]
  );
  res.json(result.rows);
});


// MongoDB aggregation pipeline
db.orders.aggregate([
  { $match: { status: "delivered" } },
  { $group: { _id: "$customer_id", total: { $sum: "$amount" } } },
```

```
  { $sort: { total: -1 } },
  { $limit: 5 }
]);
```

Expected Outcomes

Learning Outcomes

- Designed normalized relational databases using ER modeling and created and manipulated them with SQL.
- Wrote advanced SQL queries using joins, subqueries and aggregation, and worked with MongoDB documents and aggregation pipelines.
- Applied ETL and data transformation techniques and built dashboards with data models, relationships and DAX in Power BI.
- Integrated databases with Node.js/Express and FastAPI/SQLAlchemy applications through RESTful CRUD APIs.
- Applied indexing, query optimization, access control, SQL injection prevention and backup strategies to improve performance and security.
- Understood scalability and high-availability concepts and reviewed real-world case studies from e-commerce, social media and financial services.
- Developed an end-to-end, data-driven business analytics dashboard as the capstone project.

Conclusion

The internship provided a complete and practical view of database management systems, from designing a relational schema to delivering business insights through a dashboard. By working with both SQL and NoSQL databases, integrating them with backend APIs, and studying performance, security and scalability, the internship strengthened both the technical and professional skills required for data-centric software roles.

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